



The 4 families of derivative products

Forwards, futures, options, swaps: what they're for, who uses them, and what really sets them apart.

PAYOFF AT MATURITY (LONG POSITION)

Forward / Future: $S_T - K$

Call: $\max(S_T - K, 0)$

Put: $\max(K - S_T, 0)$

FORWARDS

- Over-the-counter (OTC) contract, tailor-made between 2 parties.
- No exchange of cash flows before maturity.
- Counterparty risk, not guaranteed by a clearing house.

FUTURES

- Standardized version of a forward, listed on an exchange (CME, Eurex...).
- Daily margin call (mark-to-market).
- Clearing house makes counterparty risk nearly zero.

OPTIONS

- A right, not an obligation, to buy (call) or sell (put) at K .
- Premium paid upfront by the buyer.
- Asymmetric payoff: loss capped at the premium, potentially unlimited gain.

SWAPS

- Periodic exchange of cash flows between 2 counterparties.
- IRS (fixed ↔ floating rate), cross-currency swap, CDS (credit).
- OTC, long maturities, core of ALM/rate-hedging strategies.

Who uses what?

CORPORATES

FX forwards & rate swaps to hedge debt and foreign-currency invoices.

ASSET MANAGERS

Futures to adjust exposure without touching the physical portfolio.

INSURERS

Options to protect the balance sheet against equity market declines.

BANKS

Swaps to transform and steer the balance sheet's interest rate risk (ALM).

KEY TAKEAWAYS

1. Forward vs Future: same payoff logic, but the future adds standardization + margin calls, which removes counterparty risk.
2. Options are the only "classic" derivatives with an asymmetric payoff: that's exactly why a premium is paid.
3. By notional volume, interest rate swaps dominate the OTC derivatives market — far more than options or forwards.



Comparison table

CRITERION	FORWARD	FUTURE	OPTION	SWAP
Market	OTC	Listed (exchange)	OTC or listed	OTC
Standardization	Tailor-made	Standardized	Standardized (listed) or tailor-made (OTC)	Tailor-made
Upfront payment	None	Margin deposit	Premium (buyer)	None (at inception)
Counterparty risk	High (bilateral)	Near zero (clearing house)	Depends on seller/platform	High (bilateral, unless collateralized)
Typical use case	FX hedging for a corporate	Speculation/hedging on indices, rates, commo	Asymmetric hedging, leverage	ALM management, debt transformation

Real-world examples

FX FORWARD

A French exporter sells a 6-month EUR/USD forward to lock in its exchange rate on a dollar-denominated invoice.

INDEX FUTURE

A fund manager buys E-mini S&P 500 futures to quickly adjust equity exposure without selling the underlying holdings.

HEDGING OPTION

An insurer buys equity index puts to protect its balance sheet against a market downturn (hedging life/death guarantees).

INTEREST RATE SWAP (IRS)

A bank exchanges a floating rate for a fixed rate to stabilize the cost of its long-term liabilities — core of ALM management.

Quick glossary

NOTIONAL

Reference amount used to compute cash flows — rarely exchanged as principal.

STRIKE (K)

Exercise price fixed in the contract, used as the reference for the payoff.

MARK-TO-MARKET

Revaluation of a contract at its current market value, often daily.

MARGIN CALL

Collateral payment required when a contract's value moves unfavorably.

CLEARING HOUSE

Intermediary that guarantees execution of listed contracts (futures, some options).

ISDA MASTER AGREEMENT

Standard contract governing an OTC derivatives relationship between two counterparties.

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